

A Review of New Approaches for Dynamic Vehicle Routing Problem

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Abstract—Modern logistics receives increasing attention for planning and scheduling operations of transport systems that have to be resource efficient and environmentally sustainable. This paper reviews the state-of-art of one of the main problems describing logistic issues, i.e., the Dynamic Vehicle Routing Problem (DVRP). In particular, DVRP consists in assigning routes at minimal costs, to a fleet of vehicles in order to serve a set of customers by also considering the information evolution (dynamism) and uncertainty (stochastic knowledge). Moreover, we point out how the last decade development of the Information and Communications Technology has made possible to address new real time decision making strategies. The most recent contributions are classified and discussed and the open problems are enlightened.

I. INTRODUCTION

Modern logistics recently received increasing attention from governments, business organizations and researchers in order to achieve planning, scheduling and transport systems that are responsive, resource efficient, climate and environmentally friendly. The application of automation and modern Information and Communication Technology (ICT) tools in logistics is motivated by the fact that current production and distribution logistics strategies are not sustainable in the long term and the increase in city population introduces new traffic and transportation problems in particular for waste collection and post delivery.

The culmination of all the above factors calls for new formulations of the classical Vehicle Routing Problem (VRP) that consists in assigning routes, at minimal cost, to a fleet of vehicles in order to serve a set of customers with known demands. Typically, each vehicle departs from the depot, serves a subset of customers compatibly with its capacity, and finally returns to the depot. In order to solve real-world applications, a great number of VRP variants has been introduced and intensively studied in the literature. A scheme about taxonomy is available in [1], with respect to scenario characteristics and problem physical characteristics, while short descriptions about variants can be found in [2].

The classical and *static* VRP deals with a *deterministic* operational environment where all information is well known

before optimization (at planning time) and remains static during the execution of the routing plan [2].

Actually, in the real-world, information may be affected by uncertainty and may be not always completely available because of unexpected or uncertain events, such as ongoing customer requests, actual amount of demand, breakdown of vehicles and traffic congestion. Thus, following [3] and [4], real-world applications have to consider two important dimensions of the information for the VRP: information *evolution* (static or dynamic) and information *quality* (deterministic or stochastic). Combinations of those dimensions identify four types of vehicle routing problems: static-deterministic, static-stochastic, dynamic-deterministic and dynamic-stochastic.

This paper focuses on the Dynamic VRP (DVRP) that includes dynamic-deterministic and dynamic-stochastic information. In DVRP the so called *source of dynamism* is an unknown input revealed dynamically during the execution of the routes. In dynamic-stochastic problems, stochastic knowledge on the source of dynamism is available and exploited to improve the solution. Together with real-world requirements, recent developments in ICT have accelerated the research on the DVRP [5] by stressing challenges and promising developments in automation and decision making areas.

In order to enlighten the promising perspectives in the automation area related to logistics management problems, this paper presents a review about the recent contributions in the DVRP solutions. Starting from the review paper by Pillac *et al.* [3], we discuss some of the recent advances in DVRP and new paradigms such as distributed approaches.

The paper is organized as follows. Section II introduces the DVRP with its peculiarities and challenges. Section III shows some applications, while Section IV classifies reviewed papers by solution approaches. Finally, Section V discusses the concluding remarks.

II. DYNAMIC VEHICLE ROUTING PROBLEMS

A. The DRVP Definition

According to Psaraftis [4], a VRP is *dynamic* if information (input data) on the problem is made known to the decision maker, or is updated, *concurrently* with the determination of the routes. As a consequence, a continuous reoptimization process is needed and two concurrent levels have to be considered in DVRP [6] [7]: i) a planning level, which involves optimization and reoptimization; ii) an execution level, which involves the realization of the (under construction) plan in the dynamic environment.

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As reported in [3], the most studied source of dynamism is the online arrival of new requests during the operations. Other source of dynamism, related to customers, are the variation of demands, [8], [9] or the cancellation of orders [10]. However, other sources are receiving attention such as travel times [11], [12], [13], service time at customers [13], and breakdown of vehicles [7]. In Table I the reviewed articles are classified in according to the source of dynamism and its quality of information. In particular, *cyclic* means that the pattern of dynamic changes is cyclic. References in bold represent papers that consider more than one source of dynamism.

TABLE I
SOURCES OF DYNAMISM

Source of dynamism	Quality of information	References
new requests (new customers)	deterministic	[14] [15] [6] [16] [17] [10] [18] [19] [20] [21] [7] [22] [23] [24] [25]
	stochastic	[26] [27] [28] [29] [9] [30] [31]
demand variation (same customer)	deterministic	[8] [32]
	cyclic	[8]
	stochastic	[9]
order cancellation	deterministic	[10]
service time	stochastic	[13]
travel time	deterministic	[7] [11] [12]
	stochastic	[13]
breakdown	deterministic	[7]

B. Challenges of the DVRP

Availability of real-time information opens the door to a series of features and problems that put the scheduler in front of a harder decision process, with more variables and less time to work out a solution. For example, in the context of ongoing arrival of requests, a first new problem arises: should the courier accept or deny the customer's request [3]? Again, most customers call for a *as soon as possible* service.

So, whereas in static VRP the objective function deals with minimizing some sort of travel cost (distances, times, etc. [33]), in DVRP the objective function may introduce other notions such as throughput (number of customers serviced per unit time), service denials, response time, [6] and customer satisfaction [28].

Another new aspect is the introduction of *diversions* into the DVRP [26], [7], [27]. According to Ferrucci *et al.* [27], vehicle en-route diversion describes a situation where a vehicle is currently traveling to a request location when a more urgent destination is assigned to it.

Moreover, response time imposes a compromise between reactivity and decision quality. For this reasons heuristics and meta-heuristics are preferred to exact methods and some form of decentralization [30], [11] have been introduced.

Finally, complexity of the problem, reactivity and ongoing fashion, may require some form of help to decision maker, so DVRP solvers are often part of Decision Support Systems [5], [10].

C. Degree of dynamism

The Degree of Dynamism (*DoD*) is the measure of how many events are received after the initial planning. The most common definition is the ratio between the number n_d of ongoing new requests and the total number n_{tot} of requests at the end of the transportation process:

$$DoD = \frac{n_d}{n_{tot}}$$

This and more complex formulations are reported in [3], taking in account also the lateness.

As a novelty, Lin *et. al* [10] introduce a variation that takes in account dynamic cancellation of orders or requests:

$$DoD = \frac{n_{cancel}}{n_{off}} \frac{n_{new}}{n_{off} - n_{cancel}}$$

where n_{cancel} is the number of canceled customer orders, n_{off} is the number of known customer orders (offline), and n_{new} is the number of new customer orders.

III. APPLICATIONS

DVRP assumes a continuous communication between the dispatcher and the vehicles (or among logistics entities), in order to monitor execution of routes and implement real-time control strategies, i.e. fast re-optimization of the routing plan. As a consequence, recent advances in ICT and on-board equipment, such as Global Positioning System (GPS), have increased interest in DVRP. In particular, Intelligent Transport Systems (ITS) are able to collect and process geographically spread real-time data (vehicle positions and status, third party information about traffic, etc.) and suggest operations planning [3]. More specific systems are cited and described in [3] and [5], including traffic-simulators.

Technologies makes it possible to solve the DVRP for a wide range of real world applications, where just-in-time reaction may be crucial. This section reports a set of recent contributions divided by application fields.

A. Services

In this class of applications vehicles have to visit customers and supply them with a sort of service, generally with time windows and capacity constraints.

In this area new contributions point out the necessity of developing Decision Support Systems (DSS) devoted to manage different city services. In particular, in [34] and [35] the core of a DSS devoted to postal delivery and waste collection services are respectively presented. The two problems are modelled as VRP by a Mixed Integer Linear Programming formulation with shift and time constraints. Moreover, in order to address the large dimension of the considered networks, a heuristic algorithm based on a clustering strategy and a farthest insertion heuristic for the solution of the Traveling Salesman Problem (TSP) is proposed.

In the field of waste collection services modeled as DVRP, paper [32] points out that uncertainty about filling levels of bins may cause unexpected early saturation of vehicles. In this case, all other traveling vehicles are requested to wait

until an Ant Colony System (ACS) decides either to employ a new vehicle or to reroute an already routed one.

Other service domains fall into the area of maintenance and repair operations [13]. In addition [28] considers contexts where the customer satisfaction is accomplished after more than one visit in the same planning horizon.

B. Transport of goods

About transport of goods, the most challenging environment for DVRP is city logistics [3], [33], where traffic congestion and other events are more likely to disrupt routing plans. Furthermore, customers often require "same day delivery", calling the system for shorter reaction time to changes in customer demands and traffic conditions [33].

In order to support decision makers of a hypothetical carrier agency, [5] integrates DVRP into a DSS architecture, together with modules for acquiring real-time traffic data and forecasting travel times on the base of real-time and historical data.

In the typical urban scenario, transportation is in charge to couriers (carriers), who serve shippers and end-customers (citizens) with deliveries, pickups or both in the same planning horizon (day). City logistics tries to globally optimize costs and benefits for different stakeholders [33], so some of considered papers face multi-objective optimization [14], [15], [17]. The aim is to balance customer satisfaction with travel costs.

A more holistic objective should take in account citizens' living conditions [33], which are affected by pollution and traffic congestion. In the reviewed papers about DVRP, pollution is not explicitly considered. However, readers interested in green logistics can find information about Green VRP (GVRP) and Pollution Routing Problem (PRP) in [2].

In some papers, priority of orders is considered. It may depend on how long a pending order has been waiting for service, hence postponed orders assume maximum priority for the day after planning [26], [15], [17], [18].

Finally, as an example of not packaged goods, paper [21] concerns the optimization of the journeys of refueling trucks in an airport.

C. Transport of people

Transport of people is quite similar to VRP with pickup and delivery [31] and has been often modeled as a Dial-a-Ride Problem (DARP), where main emphasis is given to customers satisfaction and reduction of passenger inconvenience [29].

Munoz *et al.* [31] face a more dynamic environment where all requests are revealed in real time, but a continuous re-optimization process exploits demand forecasting in order to route vehicles step by step.

An earlier article by Coslovich *et al.* [36] showed a distributed approach in managing dynamic requests. In their case study, at a regular bus stop and before getting on, an unexpected customer can ask directly the driver for an unplanned alighting stop. On-board continuous and autonomous re-optimization process supports the driver in making fast decisions about accepting or denying the request.

IV. SOLUTION METHODS

Along with advances in technologies, improvements to problem-solving methods have been made in order to interact with dynamic environment and to combine fast responses with solution quality [6].

In this section, recent papers are reported by following the classification proposed in [3], also considering new distributed approaches. The mentioned solution methods are summarized in Fig. 1.

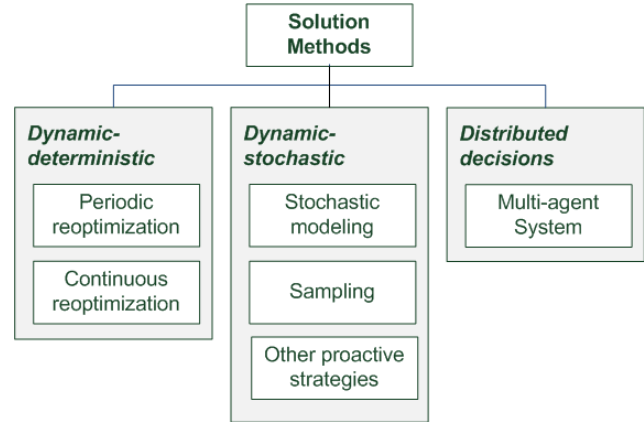


Fig. 1. Solution Methods.

A. Dynamic-deterministic routing problem

Dynamic-deterministic routing problems are solved in two main centralized approaches: a) in *periodic reoptimization*, DVRP is faced as a series of static and quite independent VRPs, each of them holds in a time interval of the workday (or execution horizon); b) in *continuous reoptimization*, an adaptive memory, most of times a pool of good solutions, is continuously updated and utilized as a feature to decide only the next destination of each vehicle [6].

1) *Periodic reoptimization*: The typical periodic reoptimization approach gives a first optimal initial routing plan (set of routes) before the beginning of the workday. After the transportation has begun, an optimization procedure periodically solves a static problem corresponding to the current state, either whenever the available data change [17], [16], or at fixed intervals of time, called decision epochs [24] or time slices [15].

For example, [15] and [14] divide the planning horizon into time slices of duration Δ , compatible with the *DoD*. At the beginning t_i of a time slice $[t_i, t_{i+1}]$ (where $t_{i+1} = t_i + \Delta$), a certain amount of time δ is spent to organize the required information and then to construct the static model. This model is solved in $[t_i + \delta, t_{i+1}]$ to find the solution S_{i+1} , which should be implemented (executed by vehicles) in the next time slice $[t_{i+1}, t_{i+2}]$. Each single VRP is solved by a Genetic Algorithm (GA).

Sarasola *et al.* [9], for example, divide the day into 25 periods and solve a static VRP in the first 5 secs of each period, by a Vehicle Neighborhood Search (VNS) employing a Clarke and Wright savings algorithm.

Okulevich *et al.* [19] use a two-phase Particle Swarm Optimization (PSO). In each time slice, a PSO assigns clients

to vehicles, then different PSOs (hybridized with 2-Opt algorithm) solve a TSP instance for each single vehicle.

The "VRP-serialization" has the advantage of employing well-known algorithms developed for static routing. On the other hand, during a time slice, vehicles perform operations decided on the base of data available at the beginning of the previous time slice. Therefore, this approach fits with relatively steady conditions.

2) *Continuous reoptimization*: In this class of approaches optimization is continuously performed throughout the time horizon while an adaptive memory stores information on good solutions. Whenever the available data change, a decision procedure aggregates the information from the memory to perform re-optimization. Examples of adaptive memory are solution pool [6], [7] and ACS with pheromone conservation [21].

Barkaoui and Gendreau [6] propose a significant example of approach based on a solution pool. At any instant t , vehicles are performing the last computed best solution, which consist of a set of planned routes, each beginning with the best current destination of the associated vehicle. The best solution is extracted from a solution pool a Genetic Algorithm continuously improves, by a search process into the solution space.

Ferrucci *et al.* [7] follow the approach of solution pool, but in a different way. More precisely, they introduce the notion of concurrency of execution and adaption along the execution time, which is divided into short anticipation horizons.

Schyns [21] proposes an adaptation of ACS in a context where the objective is the responsiveness, i.e. servicing a customer as soon as possible in its time window.

The main advantage of this class of approaches is that decisions are taken on the base of a set of already good solutions, allowing for parallelization. This approach fits with volatile conditions.

B. Dynamic-stochastic routing problem

In dynamic-stochastic routing problems, an additional stochastic knowledge about the source of dynamism can be exploited. Pillac *et al.* [3] recognize two main categories of approaches: a) in *stochastic modeling* approaches, the additional knowledge is formally included into the problem formulation; b) in *sampling* approaches, several realizations of random variable distributions (e.g. late customers requests) are performed in order to obtain the most likely scenarios in the near future and anticipate the source of dynamism.

1) *Stochastic modeling approaches*: Binart *et al.* [13] tackle a service on fields routing problem where the source of dynamism are travel times and service times (minimum and maximum values are known), whereas customers (mandatory and optional) are known. The proposed method is composed of a planning stage and an execution stage. In the planning stage a linear program (and heuristics) is solved in order to find out routes visiting all the customers (mandatory and optional), under pessimistic conditions (maximum values for travel times and services times). Pessimistic values are likely to lead to unfeasible routes (planning horizon violation),

so a repair heuristic procedure removes the farthest (from segments between mandatory customers) optional customers until feasibility is obtained. In the execution stage, dynamic programming is applied each time a vehicle finishes servicing at a customer, in order to decide if to send it to the next mandatory customer or to an optional customer (even those discarded by the repair procedure).

Muñoz-Carpintero *et al.* [31] propose a Hybrid Predictive Control approach to control a dial-a-ride system. In the proposed predictive dynamic model, the objective function depends on both current requests and expected future requests in a certain prediction horizon. The optimization problem is solved using an evolutionary algorithm each time a new request arrives, giving routes that hold only until the next new request arrival.

2) *Sampling approaches*: Sarasola *et al.* [9] face the problem of VRP with stochastic demand and dynamic requests. The particularity is that even static customers (known before the planning horizon) have stochastic demands: that is, customer demand remains uncertain until a vehicle arrives at the customer location and must serve it. At first, authors use sampling to extend a variable VNS algorithm to static-stochastic problems (Stochastic-VNS, S-VNS), then they adapt S-VNS for the dynamic setting (dynamic-stochastic problem), where the time horizon is divided into regular periods (Stochastic Dynamic VNS, SD-VNS). As authors point out, the proposed SD-VNS is good for cases with low degree of dynamism (5%-15%).

Bruni *et al.* [29] address the robust demand-responsive planning problem under uncertainty of customers by formulating it as a stochastic variant of the DARP, in order to include recourse actions (detours). By sampling the stochastic distribution of not yet revealed customers, the proposed procedure generates several different scenarios, named sub-problems, involving both the same known customers and scenario-specific customers. In the first stage, every sub-problem is solved by Tabu Search (TS). In the second stage, a coordination mechanism merges scenario solutions by evaluating a series of similarity measures. In this way a unique robust solution is found, which combines pro-activity allowed by stochastic knowledge with robustness given by the coordination mechanism.

Barkaoui *et al.* [28] tackle a generic problem in which a customer may require more visits before its satisfaction is eventually reached (multi-step repairs, visits for a diagnosis, etc.). They model this behavior by assigning to each customer a satisfaction value which is increased in according to a probabilistic function, visit by visit, until a threshold is overcome (customer is considered fully served). In this way they build a stochastic knowledge about customers that may likely require new visits in the near future. This knowledge is exploited into their solving strategy: a Hybrid Genetic Algorithm continuously generates plans including both revealed requests and potential future revisits requests (*dummy* revisits). If multiple visits are candidates for insertion in planned routes, the first visit to insert is the one with the minimum deviation of the customer average satisfaction

from the satisfaction threshold. The average is computed by simulating (sampling) plan execution over all possible histories (sequence of events) for a given solution plan.

3) *Other strategies*: Stochastic knowledge about the evolution of the source of dynamism (e.g. information about spatial and/or time distribution of late requests) allows other proactive approaches which anticipate future events (requests).

Ferrucci *et al.* [26], [27] integrate stochastic knowledge by adding so-called *dummy customers* to the set of real pending customer requests. Dummy customers guide vehicles to request-likely areas before real requests occur there. Stochastic knowledge is generated from past requests.

Request-likely areas are the base principle also for waiting strategies, in which it needs to decide when, how long and where the vehicle should wait. A vehicle can wait at the last serviced customer, or go to a strategical parking area for a waiting period (see for instance [37]).

Albareda-Sambola *et al.* [18] face a problem in which a rolling horizon is divided into sub-periods and customers have time windows covering more sub-periods. An adaptive policy exploits stochastic knowledge about next sub-period arriving requests (non-pending customers) in order to postpone some currently known customers (pending customers) to the next sub-period.

TABLE II
SOLUTION APPROACHES

<i>type of problem</i>	<i>approach</i>	<i>solution</i>	<i>references</i>
Dynamic-deterministic	Periodic reoptimization	VNS	[10] [9] [16] [17]
		GA	[14] [15]
		TS	[24]
		PSO	[19] [20]
		ACS	[8] [32]
	Continuous reoptimization	GA	[6]
		TS	[7]
		ACS	[21] [22]
	Distributed	MAS/auction	[11] [12] [23] [25]
	Dynamic-stochastic	Stoch. modeling	B&C or Lagr.
PSO or GA			[31]
Sampling		VNS	[9]
		TS	[29]
		GA	[28]
Others		TS	[26] [27]
		exact or VNS	[18]
Distributed		MAS/auction	[30]

C. Distributed approaches

The challenge of giving fast responses to frequent unexpected events, into large transportation and logistics networks, has made some researchers investigate on a paradigm shift from a centralized approach towards decentralized control of logistic processes, in order to avoid too time consuming re-computations of global routing plans [25]. This shift has been usually made by Multi-Agent Systems (MAS),

in which autonomously acting software agents represent logistic entities (carrier, vehicles, orders, etc.). Agents have the ability to interact with other agents, by the use of negotiation and communication mechanisms [23], and to take decisions.

Warden and Wojtusiak [25] consider a freight forwarding agency whose transportation process is modeled by two main classes of agents: transportation orders and trucks. More precisely, each time an incoming transport order arrives, a short-lived agent is created and becomes responsible for the re-valuation of the pending order, until it is included into a route or discarded.

Negotiation and system-wide approaches are proposed in [30] where a MAS based on interactions between shipper and carry agents improves look-ahead strategies for pickup and delivery jobs. The aims are minimizing the system-wide costs (transport costs and lateness costs) and avoiding allocations which may become unfavorable when new jobs appear. In other words, the authors study the emergent behavior of the combined system shippers-carriers by evaluating *joint* effects of their policies (auction strategies vs bidding strategies, respectively). The most interesting policy the authors propose is the Opportunity Valuation (OV) policy, applied by vehicles/carriers agents each time a shipper agent starts an auction for a new job. On the base of stochastic information on job arrival patterns and distribution of the lowest bid for various job characteristics, the OV policy (at each vehicle) employs Stochastic Dynamic Programming in order to: (i) calculate a bid price for the new job, (ii) choose an appropriate insertion position for the new job, and (iii) support so-called pro-active move decisions, i.e. moving empty in anticipation of future job requests.

Novaes *et al.* [11] consider traffic congestion as source of dynamism. Each vehicle of a fleet tries to perform an initial route but unpredictable traffic congestion can give fault, i.e., some customers are no longer reachable into the working time. The paper focuses on a MAS where vehicle agents are able to anticipate the occurrence of unperformed visits.

A more recent paper [12] completes [11] by focusing on the transference of tasks. A chief carrier agent supervises periodic auctions in which vehicles, which have detected incipient faults, can negotiate the transference of tasks to other vehicles.

Finally, the mentioned papers are listed in Table II, where: i) B&C and Lagr. are branch and cut and Lagrangian relaxation respectively; ii) bold references denote papers covering more than one approach.

V. CONCLUSIONS

This paper analyzes the recent works about the Dynamic Vehicle Routing Problem by pointing out the issues that are mainly addressed: deterministic and stochastic dynamic problems and distributed approaches. We enlighten how the new research trends are affected and incentivized by the application of the modern ICT tools in the logistic application areas.

Future researches should enhance the effort on the following matters: i) considering stochastic aspects in all the decision processes in order to capture uncertainty and forecast the demand; ii) applying parallel and distributed algorithms in the optimization approaches in order to solve large problems in real time; iii) employing Vehicle to Vehicle and Vehicle to Infrastructure communication to improve real time services. Finally, research should explicitly address DVRP involving the peculiarities of electric vehicles.

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